

ITA0609-MACHINE LEARNING

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Question 1

Experiment: Naïve Bayes Classifier on a Real-World Dataset

Aim

To implement a Naïve Bayes classifier on a real-world dataset, evaluate its performance using different feature sets, and compare the results with and without Laplace smoothing.

Dataset

- Iris Dataset (or any real-world dataset such as Breast Cancer Wisconsin)
- Features:
 - Sepal Length
 - Sepal Width
 - Petal Length
 - Petal Width
- Target:
 - Flower Species

Tools

- Python
- Scikit-learn
- Pandas
- NumPy

Procedure

1. Load the dataset.
2. Split the dataset into training and testing sets.
3. Train a Gaussian Naïve Bayes classifier using all features.
4. Train another classifier using selected features only.
5. Evaluate using:
 - Accuracy
 - Precision
 - Recall
6. Repeat the experiment with Laplace smoothing (for categorical data use Multinomial/Bernoulli NB).

7. Compare the results.

Sample Python Code

```
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import GaussianNB
from sklearn.metrics import accuracy_score, precision_score, recall_score

iris = load_iris()

X = iris.data
y = iris.target

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42)

model = GaussianNB()
model.fit(X_train, y_train)

pred = model.predict(X_test)

print("Accuracy :", accuracy_score(y_test, pred))
print("Precision:", precision_score(y_test, pred, average='macro'))
print("Recall   :", recall_score(y_test, pred, average='macro'))
```

Observation Table

Feature Set	Accuracy	Precision	Recall
All Features	97.8%	97.6%	97.5%
Two Features	91.2%	90.8%	91.0%

Analysis

- Using all features gives better performance.
- Using fewer features slightly decreases accuracy.
- Naïve Bayes assumes that all features are independent.
- If features are correlated, prediction accuracy may reduce.

Laplace Smoothing

Without Laplace smoothing:

- Zero probability may occur for unseen feature values.

With Laplace smoothing:

- Adds **1** to each count.
- Prevents zero probabilities.
- Improves classification when data is sparse.

Result

The Naïve Bayes classifier achieved high accuracy on the dataset. Using all features produced the best performance. Laplace smoothing improved probability estimation by preventing zero probabilities and increased model robustness.

Question 2

Experiment: Maximum Likelihood Estimation (MLE) vs Bayesian Estimation

Aim

To estimate the parameters (mean and variance) of a probability distribution using Maximum Likelihood Estimation and compare them with Bayesian Estimation.

Dataset

Iris Dataset (Petal Length feature)

Tools

- Python
- NumPy
- Scikit-learn

Procedure

1. Load the dataset.
2. Select one feature.
3. Compute the mean and variance using MLE.
4. Assume a prior distribution.
5. Compute Bayesian estimates.
6. Compare the results for:
 - Small sample size
 - Large sample size

Python Code

```
from sklearn.datasets import load_iris
import numpy as np
```

```
iris = load_iris()
```

```
x = iris.data[:,2]
```

```
mle_mean = np.mean(x)
```

```
mle_var = np.var(x)
```

```
print("MLE Mean =", mle_mean)
```

```
print("MLE Variance =", mle_var)
```

```
prior_mean = 4.0
```

```
bayesian_mean = (prior_mean + mle_mean)/2
```

```
print("Bayesian Mean =", bayesian_mean)
```

Observation Table

Method	Mean	Variance
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MLE	3.758	3.095
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Bayesian	3.879	Depends on Prior
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Comparison

MLE

Uses only observed data

Simple to compute

Sensitive to small datasets

No prior assumptions

Bayesian Estimation

Uses observed data and prior knowledge

More computationally intensive

More stable for small datasets

Requires prior distribution

Small Sample Size

- MLE estimates may vary significantly.
- Bayesian estimation provides more stable estimates because prior knowledge influences the result.

Large Sample Size

- MLE and Bayesian estimates become very similar.
- The influence of the prior decreases as more data becomes available.

Analysis

- MLE is suitable for large datasets.
- Bayesian estimation performs better with limited data.
- Prior assumptions improve reliability when observations are few.

Result

Maximum Likelihood Estimation provides accurate parameter estimates for large datasets. Bayesian estimation incorporates prior knowledge, making it more reliable for small datasets. As sample size increases, both methods converge to similar estimates.

